Introduction and Structures of Materials - II



MO-201 Materials Science
Odd Sem 2026-27
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Resolving Power of Microscope

- In 2D, the Abbe criterion gives the resolution d of a microscope as

$$d = 0.61 \frac{\lambda}{n sin\alpha}$$

where λ is the wavelength, n is the refractive index of the imaging medium (air, oil, vacuum), and α is half of the cone angle of light from the specimen plane accepted by the objective (half aperture angle in radians), in which case nsinα is the numerical aperture

- Numerical aperture is ~1 for light microscope, ~0.01 for electron microscope

- As the wavelength of light is 400–700 nm, the resolution of a light microscope is ~250–420 nm

Methods Enzymol. 2010 ; 482: 73–100. doi:10.1016/S0076-6879(10)82003-8

Resolving Power of Microscope

- For electron microscopes, the electron wavelength λ depends on the h^2 accelerating voltage V
- Within the classical approximation,

$$\lambda = \sqrt{\frac{h^2}{2meV}}$$

where h is Planck's constant, m is mass and e the charge of the electron

- For an accelerating voltage 100 keV, λ = 0.0037 nm
- The theoretical resolution of electron microscopes is 0.23 nm for an accelerating voltage 100 keV (λ = 0.003701nm) and 0.12 nm for 300keV (λ = 0.001969 nm), and is mainly limited by the quality of electron optics

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Crystalline and Noncrystalline States

- The number and kind of nearest neighbours, that an atom or an ion has in a solid, is nearly the same for both the crystalline and the noncrystalline forms of the solid

- However, the **noncrystalline structure does not have the long range periodicity** characteristic of the crystalline state

- In a crystal, any number of integral lattice translations would take us from an atom located at a lattice point to another identical atom located at a different lattice point; this is not true for the noncrystalline state

- As long range periodicity is the basis of diffraction effects, the noncrystalline solids do not give rise to sharp diffraction patterns like crystals

Factors promoting formation of noncrystalline structures

- **Secondary Bonding Forces**

- When primary bonds do not extend in all directions, 1D chain molecules or 2D sheet molecules are formed; such units have to be aided by secondary bonding forces to form a 3D crystal

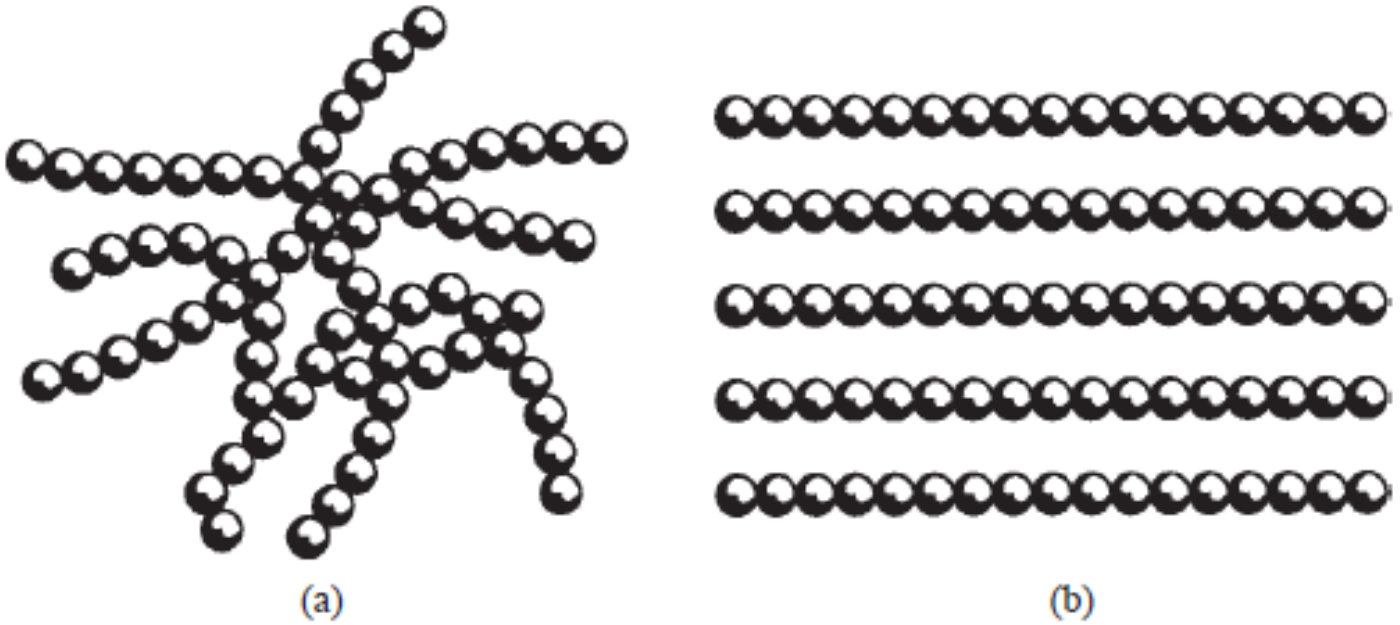
- Consider a structure consisting of long chain molecules; in the molten state, the chains persist and are like a bowl of wriggling earthworms (“wriggling” refers to the translational motion of the chains past one another)

- On cooling, if the secondary bonding forces are not strong enough to exert themselves, the chains cannot get straightened out of the tangle to become the orderly, parallel arrangement of a crystal

- The translational freedom is gradually lost during cooling, till the noncrystalline, glassy state is reached, where only the vibrational degree of freedom remains

Factors promoting formation of noncrystalline structures

- **Secondary Bonding Forces**



(a) The tangled up configuration of long chain molecules, (b) the parallel array of the chain molecules characteristic of a crystal.

Factors promoting formation of noncrystalline structures

- **Free Energy**

- The free energy of the crystalline state is always lower than that of the noncrystalline state

- Only when the free energy difference between the two states is large in magnitude, the tendency to crystallize will be strong

- Some materials form a relatively open network structure of atoms, where there is little free energy difference between an orderly array and a disorderly array of the units

- In such cases, the tendency to crystallize will be weak

Factors promoting formation of noncrystalline structures

- **Cooling Kinetics**

- A third factor that promotes the formation of noncrystalline structures is the rate of cooling from the liquid state

- Slow cooling rates allow enough time for crystallization, while fast cooling rates may prevent crystallization altogether

- The rate of cooling which is considered as slow or fast will vary widely for different materials, depending on the magnitude of the kinetic barrier

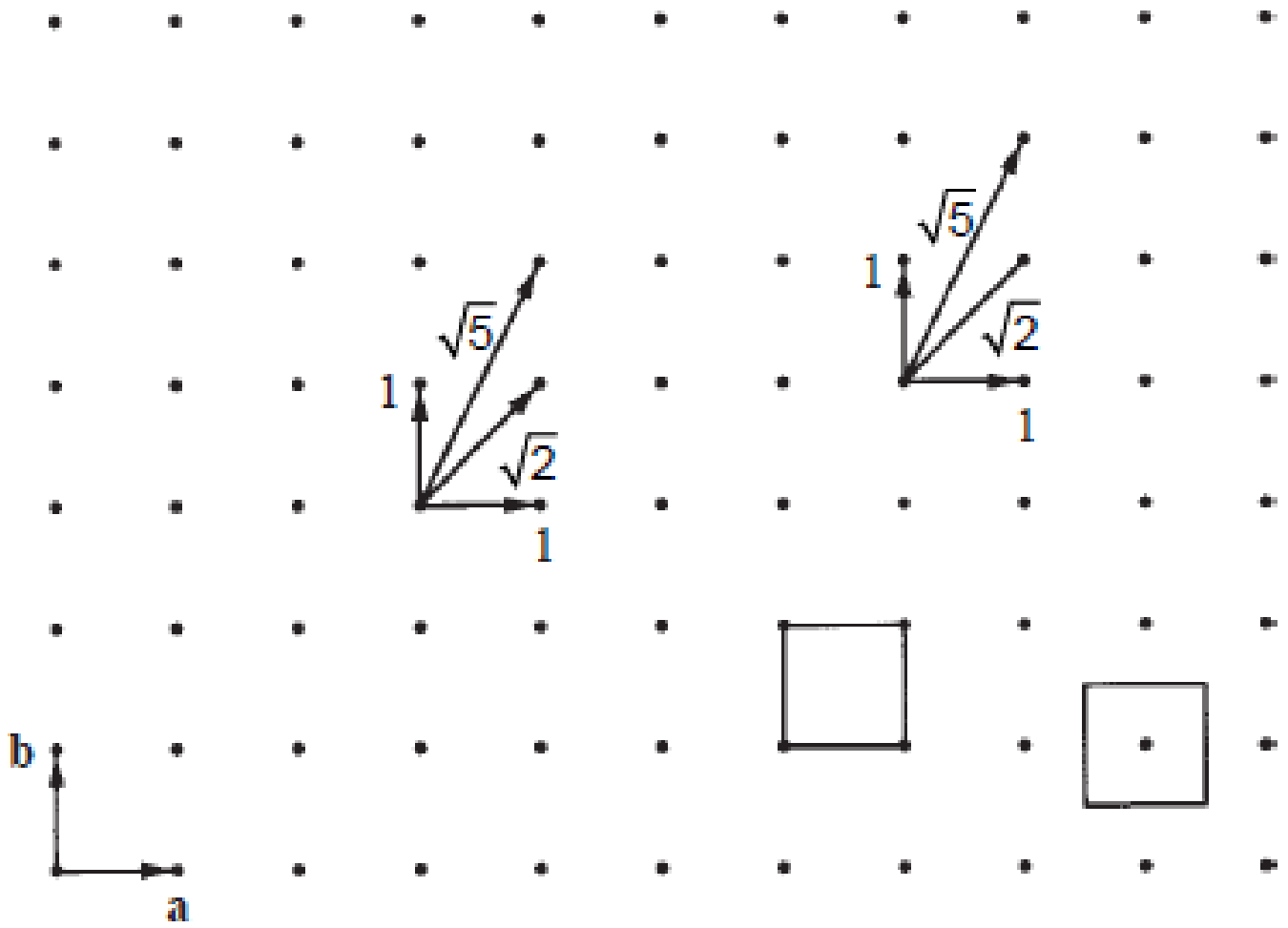
- For a metal, the cooling rate required to prevent crystallization may be as high as 10^6 K s^{-1}

- For a silicate, cooling at the rate of a fraction of K per hour may be sufficient to prevent crystallization

Crystalline vs Noncrystalline

- The crystal exhibits a sharp melting point in contrast to the noncrystalline material, which gradually softens over a range of temperature
- As a result of the regularity of the arrangement, the atoms or molecules in a crystal are more closely packed
- Hence, the crystalline form has a higher density than the noncrystalline form
- The closer packing in the crystal tends to increase the average strength of the secondary bonds present

Space Lattices



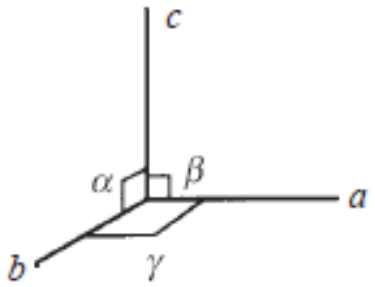
A two-dimensional square array of points gives a square lattice. Two ways of choosing a unit cell are illustrated

Bravais Lattices

I. Cubic

$a = b = c$

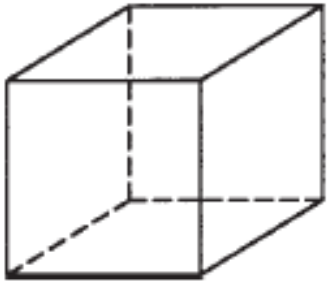
$\alpha = \beta = \gamma = 90°$



(1) Simple (Lattice points at the eight corners of the unit cell)

(2) Body centred (Points at the eight corners and at the body centre)

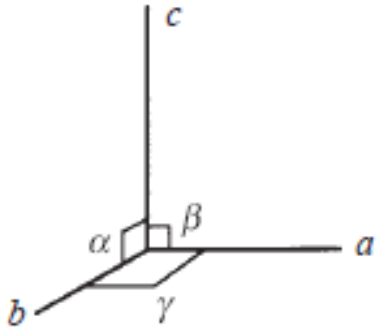
(3) Face centred (Points at the eight corners and at the six face centres)



II. Tetragonal

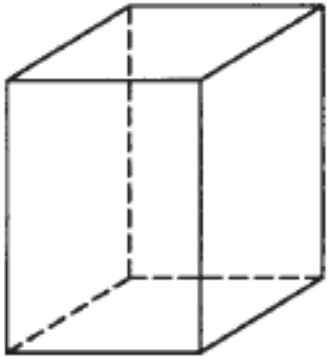
$a = b \neq c$

$\alpha = \beta = \gamma = 90°$



(4) Simple (Points at the eight corners of the unit cell)

(5) Body centred (Points at the eight corners and at the body centre)

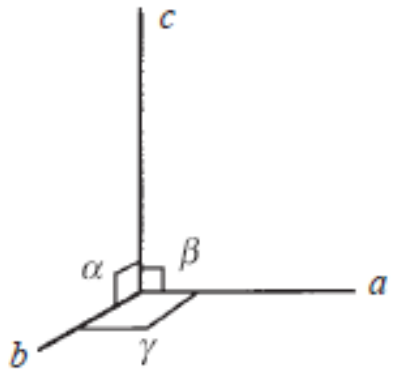


Bravais Lattices

III. Orthorhombic

$a \neq b \neq c$

$\alpha = \beta = \gamma = 90°$

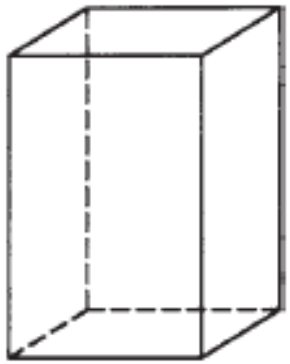


(6) Simple (Points at the eight corners of the unit cell)

(7) End centred (Also called side centred or base centred) (Points at the eight corners and at two face centres opposite to each other)

(8) Body centred (Points at the eight corners and at the body centre)

(9) Face centred (Points at the eight corners and at the six face centres)

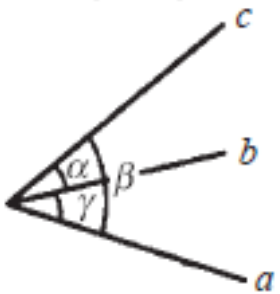


Bravais Lattices

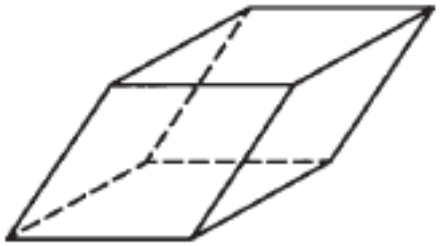
IV. Rhombohedral

$a = b = c$

$\alpha = \beta = \gamma \neq 90°$



(10) Simple (Points at the eight corners of the unit cell)

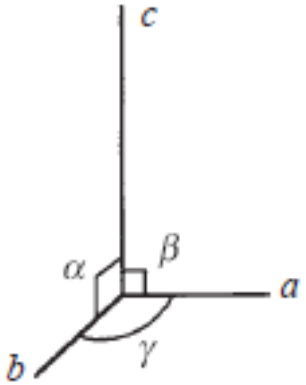


V. Hexagonal

$a = b \neq c$

$\alpha = \beta = 90°$

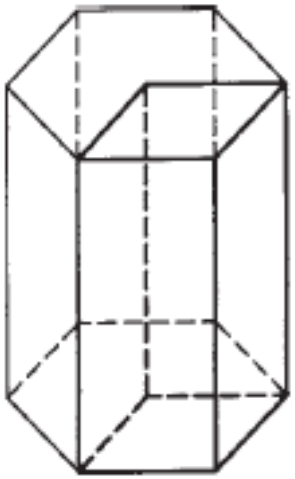
$\gamma = 120°$



(11) Simple [(i) Points at the eight corners of the unit cell outlined by thick lines

or

(ii) Points at the twelve corners of the hexagonal prism and at the centres of the two hexagonal faces]

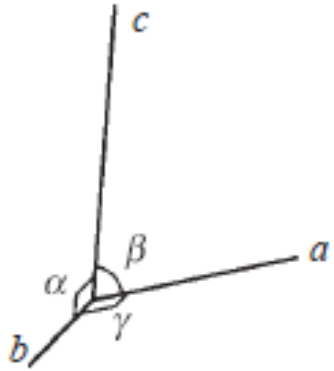


Bravais Lattices

VI. Monoclinic

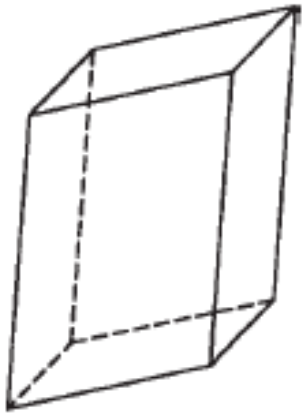
$a \neq b \neq c$

$\alpha = \gamma = 90° \neq \beta$



(12) Simple (Points at the eight corners of the unit cell)

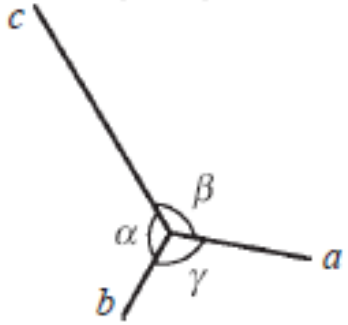
(13) End centred (Points at the eight corners and at two face centres opposite to each other)



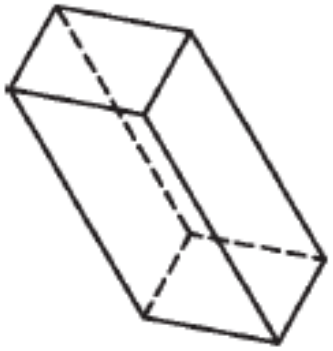
VII. Triclinic

$a \neq b \neq c$

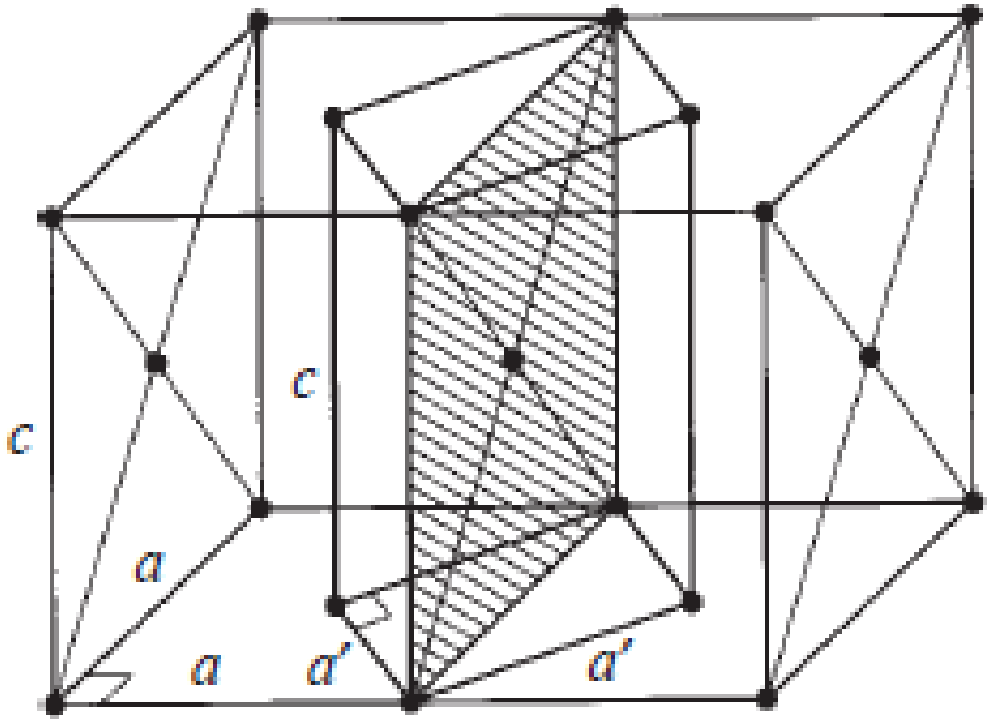
$\alpha \neq \beta \neq \gamma \neq 90°$



(14) Simple (Points at the eight corners of the unit cell)



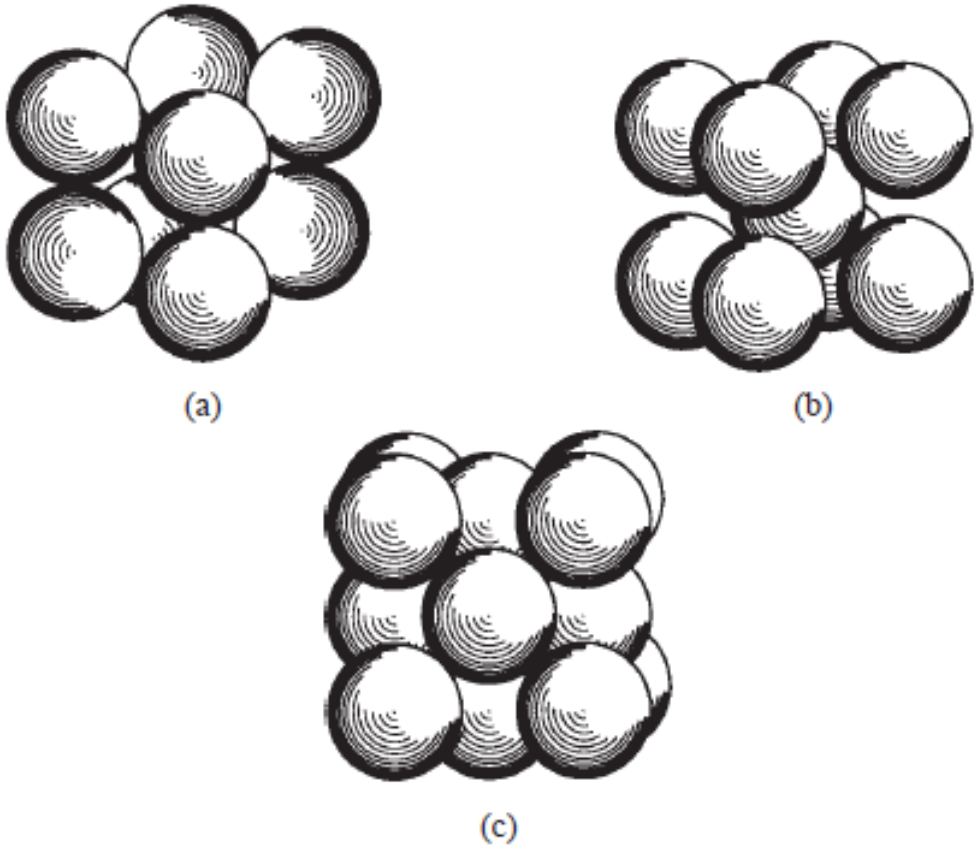
Bravais Lattices



An array of lattice points that fit in an FCT unit cell should be represented by a BCT cell (outlined by thick lines inside two adjacent FCT cells). [For clarity, the face centred atoms on the front and back faces are omitted.]

Crystal Structure

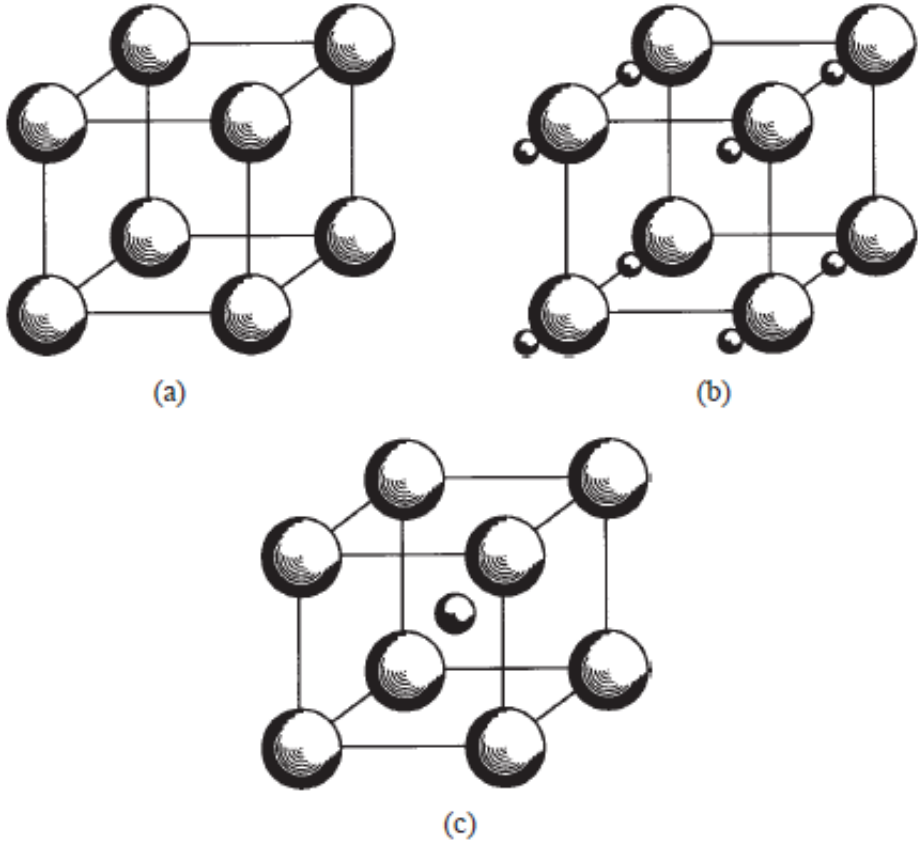
Crystal structure = Lattice + Basis



Unit cells of monoatomic crystals of (a) simple cubic (SC), (b) body centred cubic (BCC), and (c) face centred cubic (FCC) structure.

Crystal Structure

Crystal structure = Lattice + Basis



Three different crystal structures referred to the same simple cubic lattice. [For clarity, neighbouring atoms or molecules are shown separated.]

References

- Materials Science and Engineering, A First Course by V. Raghavan
- Fundamentals of Materials Science and Engineering, W.D. Callister
- Try these free interactive online crystal visualizers
  - https://www.solidstate3d.com/
  - https://www.crystallify.com/
  - https://crystalwalk.herokuapp.com/
  - https://crystals.symotter.org/viztools